1.50 Proof by Induction: Inequalities and Non-standard Proofs

Worked Solutions

1 a)

$$2 \times 2^k = 2^{k+1}.$$

1 b)

$$-3 \times 3^k = -3^{k+1}.$$

1 c)

$$a \times a^k = a^{k+1}.$$

2 a)

$$(1+a) \times (1+a)^k = (1+a)^{k+1}.$$

2 b)

$$(1-a) \times (1-a)^k = (1-a)^{k+1}.$$

3 a)

$$3 \times 2! = 3! = 6.$$

3 b)

$$12 \times 11! = 12!.$$

3 c)

$$(k+1)k! = (k+1)!.$$

4)

Prove true for $n = 3$: $2^3 = 8$ and $2 \times 3 = 6$ and since $8 \geq 6$ so true for $n = 3$.

Assume true for $n = k$: $2^k > 2k$.

Prove true for $n = k + 1$:

$$2^{k+1} = 2 \times 2^k.$$

So since $2^k > 2k$, $2^{k+1} > 2 \times 2k$.

$$2^{k+1} > 4k = 2k + 2k.$$

Since $k \geq 3$, we know that $2k \geq 6$.

$$2^{k+1} > 2k + 6 > 2k + 2 = 2(k + 1),$$

which is $2^n > 2n$ with $n = k + 1$ substituted in.

As the statement is true for $n = 3$, and if the statement is true for $n = k$ then the statement is true for $n = k + 1$, so the statement is true for all integers $n \geq 3$.

5)

Prove true for $n = 2$: $4^2 = 16$ and $3 \times 2 + 7 = 13$ and since $16 > 13$ so true for $n = 2$.

Assume true for $n = k$: $4^k > 3k + 7$.

Prove true for $n = k + 1$:

$$4^{k+1} = 4 \times 4^k.$$

So since $4^k > 3k + 7$, then $4 \times 4^k > 4(3k + 7) = 12k + 28$. As $k \geq 2$, $12k + 28 > 3k + 10$.

$$4^{k+1} > 3k + 10 = 3k + 3 + 7 = 3(k + 1) + 7,$$

which is $4^n > 3n + 7$ with $n = k + 1$ substituted in.

As the statement is true for $n = 2$, and if the statement is true for $n = k$ then the statement is true for $n = k + 1$, so the statement is true for all integers $n \geq 2$.

6)

Prove true for $n = 5$: $5^2 = 25$ and $2^5 = 32$ and since $25 < 32$ true for $n = 5$.

Assume true for $n = k$: $k^2 < 2^k$.

Prove true for $n = k + 1$:

$$2k^2 < 2 \times 2^k, \qquad 2k^2 < 2^{k+1}, \qquad k^2 + k^2 < 2^{k+1}.$$

Since $k > 4$ then $k^2 > 4k$.

$$k^2 + 4k < 2^{k+1}, \qquad k^2 + 2k + 2k < 2^{k+1}.$$

Since $k > 4$ then $2k > 8$.

$$k^2 + 2k + 8 < 2^{k+1}, \qquad (k + 1)^2 + 7 < 2^{k+1}.$$

So it holds that $(k + 1)^2 < 2^{k+1}$, which is $n^2 < 2^n$ with $n = k + 1$ substituted in.

As the statement is true for $n = 5$, and if the statement is true for $n = k$ then the statement is true for $n = k + 1$, so the statement is true for all integers $n \geq 5$.

7 a)

$3^3 = 27$ and $3^3 = 27$ so it doesn't hold when $n = 3$. $a = 3$.

7 b)

When $n = 1$ then $1^3 = 1$ and $3^1 = 3$ so true for $n = 1$. When $n = 2$ then $2^3 = 8$ and $3^2 = 9$ so true for $n = 1$.

7 c)

Prove true for $n = 4$: $4^3 = 64$ and $3^4 = 81$ and since $64 < 81$ true for $n = 4$.

Assume true for $n = k$: $k^3 < 3^k$.

Prove true for $n = k + 1$:

$$3k^3 < 3 \times 3^k, \qquad 3k^3 < 3^{k+1}, \qquad k^3 + 2k^3 < 3^{k+1}.$$

Since $k > 3$, $2k^3 > 6k^2$.

$$k^3 + 6k^2 < 3^{k+1}, \qquad k^3 + 3k^2 + 3k^2 < 3^{k+1}.$$

Since $k > 3$, $3k^2 > 9k$.

$$k^3 + 3k^2 + 9k < 3^{k+1}, \qquad k^3 + 3k^2 + 3k + 6k < 3^{k+1}.$$

Since $k > 3$, $6k > 18$.

$$k^3 + 3k^2 + 3k + 18 < 3^{k+1}, \qquad (k+1)^3 + 17 < 3^{k+1}, \qquad (k+1)^3 < 3^{k+1},$$

which is $n^3 < 3^n$ with $n = k + 1$ substituted in.

True for $n = 1$ and $n = 2$. As the statement is true for $n = 4$, and if the statement is true for $n = k$ then the statement is true for $n = k + 1$, so the statement is true for all integers $n \geq 4$.

8)

Prove true for $n = 4$: $4! = 24$ and $2^4 = 16$ since $24 > 16$, so true for $n = 4$.

Assume true for $n = k$: $k! > 2^k$.

Prove true for $n = k + 1$:

$$(k+1)! = (k+1)k!.$$

Since $k! > 2^k$, $(k+1)k! > (k+1)2^k$. Since $k > 1$ then $k + 1 > 2$, so $(k+1)2^k > 2 \times 2^k$.

$$(k+1)! > 2^{k+1},$$

which is $n! > 2^n$ with $n = k + 1$ substituted in.

As the statement is true for $n = 4$, and if the statement is true for $n = k$ then the statement is true for $n = k + 1$, so the statement is true for all integers $n \geq 4$.

9)

Prove true for $n = 5$: $5! = 120$ and $10 \times 5 = 50$, so true for $n = 5$.

Assume true for $n = k$: $k! > 10k$.

Prove true for $n = k + 1$: Since $k! > 10k$,

$$(k+1)k! > (k+1)10k, \qquad (k+1)! > 10k^2 + 10k.$$

Since $k \geq 5$, $10k^2 \geq 50k$, so

$$(k+1)! > 50k + 10k > 10k + 50k.$$

Since $k \geq 5$, $50k \geq 250$, so

$$(k+1)! > 10k + 250 > 10k + 10 = 10(k+1),$$

which is $n! > 10n$ with $n = k + 1$ substituted in.

As the statement is true for $n = 5$, and if the statement is true for $n = k$ then the statement is true for $n = k + 1$, so the statement is true for all integers $n \geq 5$.

10)

Prove true for $n = 5$: $3^5 = 243$ and $(5+1)^3 = 216$, so true for $n = 5$.

Assume true for $n = k$: $3^k > (k+1)^3$.

Prove true for $n = k+1$:

$$3^{k+1} = 3 \times 3^k > 3(k+1)^3,$$

so

$$3^{k+1} > 3(k+1)^3 = 3(k^3 + 3k^2 + 3k + 1)$$
$$= 3k^3 + 9k^2 + 9k + 3 = k^3 + 2k^3 + 9k^2 + 9k + 3.$$

Since $k > 4$, $2k^3 > 8k^2$.

$$3^{k+1} > k^3 + 8k^2 + 9k^2 + 9k + 3 = k^3 + 17k^2 + 9k + 3$$
$$= k^3 + 3k^2 + 14k^2 + 9k + 3.$$

Since $k > 4$, $14k^2 > 56k$.

$$3^{k+1} > k^3 + 3k^2 + 56k + 9k + 3 = k^3 + 3k^2 + 65k + 3$$
$$= k^3 + 3k^2 + 6k + 59k + 3.$$

Since $k > 4$, $59k > 236$.

$$3^{k+1} > k^3 + 3k^2 + 6k + 236 + 3 = k^3 + 3k^2 + 6k + 239$$
$$= k^3 + 3k^2 + 6k + 8 + 231 = (k+2)^3 + 231$$
$$> (k+2)^3,$$

which is $3^n > (n+1)^3$ with $n = k+1$ substituted in.

As the statement is true for $n = 5$, and if the statement is true for $n = k$ then the statement is true for $n = k+1$, so the statement is true for all integers $n \geq 5$.

11)

Prove true for $n = 4$: $4! = 24$ and $4^2 + 4 = 20$, so true for $n = 4$.

Assume true for $n = k$: $k! > k^2 + k$.

Prove true for $n = k+1$: Since $k! > k^2 + k$,

$$(k+1)k! > (k+1)(k^2 + k), \qquad (k+1)! > k^3 + 2k^2 + k.$$

Since $k \geq 4$, $k^3 \geq 4k^2$, so

$$(k+1)! > 4k^2 + 2k^2 + k = k^2 + 5k^2 + k.$$

Since $k \geq 4$, $5k^2 \geq 20k$, so

$$(k+1)! > k^2 + 20k + k = k^2 + 3k + 18k.$$

Since $k \geq 4$, $18k \geq 72$, so

$$(k+1)! > k^2 + 3k + 72 > k^2 + 3k + 2$$
$$= k^2 + 2k + 1 + k + 1 = (k+1)^2 + (k+1),$$

which is $n! > n^2 + n$ with $n = k+1$ substituted in.

As the statement is true for $n = 4$, and if the statement is true for $n = k$ then the statement is true for $n = k+1$, so the statement is true for all integers $n \geq 4$.

12)

Prove true for $n = 4$:

$$\left(1 + \frac{1}{4}\right)^3 = \frac{125}{64} < 2,$$

so true for $n = 4$.

Assume true for $n = k$: $2 > \left(1 + \frac{1}{k}\right)^3$.

Prove true for $n = k + 1$: Since $\frac{1}{k+1} < \frac{1}{k}$, then $1 + \frac{1}{k} > 1 + \frac{1}{k+1}$. Since k is positive,

$$\left(1 + \frac{1}{k}\right)^3 > \left(1 + \frac{1}{k+1}\right)^3.$$

So

$$2 > \left(1 + \frac{1}{k}\right)^3 > \left(1 + \frac{1}{k+1}\right)^3,$$

therefore $2 > \left(1 + \frac{1}{k+1}\right)^3$, which is $2 > \left(1 + \frac{1}{n}\right)^3$ with $n = k + 1$ substituted in.

As the statement is true for $n = 4$, and if the statement is true for $n = k$ then the statement is true for $n = k + 1$, so the statement is true for all integers $n \geq 4$.

13)

Prove true for $n = 2$:

$$(1 + a)^2 = 1 + 2a + a^2 \Rightarrow (1 + a)^2 > 1 + 2a,$$

so true for $n = 2$.

Assume true for $n = k$: $(1 + a)^k > 1 + ak$.

Prove true for $n = k + 1$:

$$(1 + a)^{k+1} = (1 + a)(1 + a)^k.$$

Since $a > -1$, multiply the assumption through by $(1 + a)$:

$$(1 + a)(1 + a)^k > (1 + a)(1 + ak)$$

$$(1 + a)^{k+1} > 1 + a + ak + a^2k = 1 + a(k + 1) + a^2k.$$

Since $a^2k \geq 0$,

$$(1 + a)^{k+1} > 1 + a(k + 1),$$

which is $(1 + a)^n > 1 + an$ with $n = k + 1$ substituted in.

As the statement is true for $n = 1$, and if the statement is true for $n = k$ then the statement is true for $n = k + 1$, so the statement is true for all integers $n \geq 1$.

14)

Prove true for $n = 2$: $2! = 2$ and $2^2 = 4$, so true for $n = 2$.

Assume true for $n = k$: $k! < k^k$.

Prove true for $n = k + 1$: Since $k! < k^k$,

$$(k + 1)k! < (k + 1)k^k, \qquad (k + 1)! < (k + 1)k^k.$$

Since $(k + 1)k^k < (k + 1)(k + 1)^k = (k + 1)^{k+1}$,

$$(k + 1)! < (k + 1)^{k+1},$$

which is $n! < n^n$ with $n = k + 1$ substituted in.

As the statement is true for $n = 2$, and if the statement is true for $n = k$ then the statement is true for $n = k + 1$, so the statement is true for all integers $n \geq 2$.

15)

Prove true for $n = 1$:
LHS: ABA^{-1} and RHS: AB^1A^{-1} so true for $n = 1$.

Assume true for $n = k$: $(ABA^{-1})^k = AB^kA^{-1}$.

Prove true for $n = k + 1$:

$$(ABA^{-1})^{k+1} = (ABA^{-1})^k(ABA^{-1})$$

$$= AB^kA^{-1}(ABA^{-1}) = AB^kBA^{-1} = AB^{k+1}A^{-1},$$

so true for $n = k + 1$.

As the statement is true for $n = 1$, and if the statement is true for $n = k$ then the statement is true for $n = k + 1$, so the statement is true for all integers $n \geq 1$.

16)

Prove true for $n = 1$:
LHS: $I + A$ and RHS: $I + 1A$ so true for $n = 1$.

Assume true for $n = k$: $I + A + A^2 + \cdots + A^k = I + kA$.

Prove true for $n = k + 1$:

$$I + A + A^2 + \cdots + A^k + A^{k+1} = I + kA + A^{k+1}.$$

But since $A^2 = A$ then $A^3 = A$ so $A^{k+1} = A$.

$$I + A + A^2 + \cdots + A^k + A^{k+1} = I + kA + A = I + (k + 1)A,$$

which is $I + nA$ with $n = k + 1$ substituted in.

As the statement is true for $n = 1$, and if the statement is true for $n = k$ then the statement is true for $n = k + 1$, so the statement is true for all integers $n \geq 1$.

17)

Prove true for $n = 1$:
LHS: $\det(A_1) = \det(A)$;
RHS: $(\det(A))^1 = \det(A)$ so true for $n = 1$.

Assume true for $n = k$: $\det(A_k) = (\det(A))^k$.

Prove true for $n = k + 1$:

$$\det(A_{k+1}) = \det(A \times A_k) = \det(A)\det(A_k)$$

$$= \det(A)(\det(A))^k = (\det(A))^{k+1},$$

so true for $n = k + 1$.

As the statement is true for $n = 1$, and if the statement is true for $n = k$ then the statement is true for $n = k + 1$, so the statement is true for all integers $n \geq 1$.

18)

Prove true for $n = 1$:

$$\frac{d}{dx}(x^a) = ax^{a-1}, \qquad \frac{a!}{(a-1)!}x^{a-1} = \frac{(a-1)!a}{(a-1)!}x^{a-1} = ax^{a-1},$$

so true for $n = 1$.

Assume true for $n = k$: $\dfrac{d^k}{dx^k}(x^a) = \dfrac{a!}{(a-k)!}x^{a-k}$.

Prove true for $n = k + 1$:

$$\frac{d^{k+1}}{dx^{k+1}}(x^a) = \frac{d}{dx}\left(\frac{d^k}{dx^k}(x^a)\right)$$

$$= \frac{d}{dx}\left(\frac{a!}{(a-k)!}x^{a-k}\right) = \frac{a!}{(a-k)!}(a-k)x^{a-k-1}$$

$$= \frac{a!}{(a-k)(a-k-1)!}(a-k)x^{a-k-1}$$

$$= \frac{a!}{(a-k-1)!}x^{a-k-1} = \frac{a!}{(a-(k+1))!}x^{a-(k+1)},$$

which is $\frac{d^n}{dx^n}(x^a) = \frac{a!}{(a-n)!}x^{a-n}$ with $n = k + 1$ substituted in.

As the statement is true for $n = 1$, and if the statement is true for $n = k$ then the statement is true for $n = k + 1$, so the statement is true for all integers $n \geq 1$.

19)

Prove true for $n = 0$:

$$\frac{0^2 + 0 + 2}{2} = 1,$$

which means there is an undivided plane, so true for $n = 0$.

Assume true for $n = k$: the plane is divided into $\frac{k^2+k+2}{2}$ regions.

Prove true for $n = k + 1$:
Since no two lines are parallel and no three meet at one point, the new line intersects all the existing k lines at distinct points. These k lines divide the new line into $k + 1$ segments. Each segment adds one new region to the plane:

$$\frac{k^2 + k + 2}{2} + (k + 1) = \frac{k^2 + k + 2}{2} + \frac{2(k + 1)}{2}$$

$$= \frac{k^2 + 3k + 4}{2} = \frac{k^2 + 2k + 1 + k + 1 + 2}{2}$$

$$= \frac{(k + 1)^2 + (k + 1) + 2}{2}.$$

So this is true for $n = k + 1$.

As the statement is true for $n = 0$, and if the statement is true for $n = k$ then the statement is true for $n = k + 1$, so the statement is true for all integers $n \geq 0$.